

Prize Question
of the Physical-Mathematical Class of the Royal Prussian Academy of
Sciences, for the Jubilee of the Accession of King Frederick II, for the year
1844

Announced on the anniversary of Frederick II's accession, 31 May 1840.

The theorem which science owes to ABEL, remarkable alike for its generality and its simplicity, seems to contain the germ of a complete theory of all integrals whose element is an algebraic function of the variable. In the simplest forms of this function Abel's theorem passes into the long-known fundamental equations of the trigonometric and elliptic functions; and, from the extent and importance that the theory of these two kinds of transcendents has attained through the repeated efforts of mathematicians, one may already infer, with great probability, the future significance of the general theory which Abel prepared by his discovery. What has so far been accomplished on the foundation laid by him, chiefly by LEGENDRE, JACOBI, and RICHELLOT, may be regarded as a first important beginning of an extensive discipline, which will undoubtedly furnish analysts with material for the most comprehensive investigations for a long time to come. For these investigations the analogy that the subject presents with transcendents of a similar but simpler nature, already so widely explored, appears to offer a powerful aid; and one may promise oneself all the greater success from using it, since the complete transformation that the theory of elliptic functions has undergone in recent times has itself made that theory more similar to the earlier-developed doctrine of circular functions.

Although the enlargement and enrichment of integral calculus just mentioned, like all more significant analytical discoveries, did not arise from a single thought, but from the cooperation of several mutually supporting thoughts, one of them nevertheless appears to deserve the greatest importance, because it has been more effective than any other in producing this transformation and intimately permeates all parts of the new theory. Earlier workers on this subject regarded the elliptic integral as a function of its amplitude; the new mode of consideration proceeds essentially from the opposite point of view and treats the amplitude, or rather certain trigonometric combinations of it, as functions of the integral, just as one had earlier reached the most important properties of the transcendents depending on the circle by regarding sine and cosine as functions of the arc, and not the arc as a function of them. The numerous and brilliant results that have followed from this new treatment make it highly desirable that the same mode of consideration be applied to the more complicated transcendents introduced into science by ABEL and whose fundamental properties he established. JACOBI has already taken an important step in this direction by showing that the inverse functions corresponding to Abelian integrals contain two or more variables and possess the remarkable property of being fourfold or multiply periodic. This result sheds an entirely new light on the nature of these transcendents, but at the same time reveals the full extent of the difficulties that stand in the way of a complete representation of these inverse functions, and that must be overcome if the theory of Abelian transcendents is to be brought to the same stage of development as that which the theory of elliptic functions has already reached.

Convinced of the advantages that analysis must gain from the further development of this theory, the Royal Academy, being prompted by the memorial celebration of Frederick the Second's accession to the throne to open an extraordinary prize competition, believes that it makes a choice worthy of this celebration in submitting this subject to mathematicians for treatment. It therefore asks for:

A detailed investigation of Abelian integrals, and especially of the functions of two or more variables which are to be regarded as their inverse functions.

The Academy abstains from any closer determination of the scope to be given to the treatment of the subject, since only the work itself can decide whether Abelian integrals can already be investigated with success in their full generality, or whether one must first restrict oneself to special classes of them, perhaps even to the class that follows immediately upon the elliptic functions.